

Data Job Roles Interview Questions

1. Tell Me About Yourself? / Walk me through your resume?

Hello, my name is Shivam Shere. I completed my Bachelor of Engineering in Artificial Intelligence and Data Science from Mumbai University in 2025.

Currently, I am working as a Data Analytics Intern at Tata Motors Subsidiary Tata Motors Digital.AI Labs in Pune, where I work with the Central Analytics team. In this role, I Assisted Senior Data Engineers and Senior Power BI Developers to converting business requirements into data-driven solutions.

My responsibilities include, validating and monitoring automated data pipelines, processing IoT Sensors and production data, developing Power BI dashboards that supports business decision-making.

Apart from my internship, I have worked on multiple personal projects like a Finance and Customer Segment Analytics Dashboard built using Power BI Desktop, Power BI Service and Microsoft Fabric. Where I performed ETL Process, designed a Star Schema data model, created Time Intelligence based DAX measures, and published reports to Power BI Service.

My key skills include SQL, Power BI, DAX, Python, Microsoft Fabric / Databricks, Azure Cloud, PySpark, Spark SQL, along with Data Engineering and Warehousing Concepts.

I enjoy solving data-related problems and learning new technologies, and now looking for an opportunity where I can contribute as a Data Professional, where I can apply my Skills to solve real world business problems, build my Career, and contribute to the growth of organization.

2. Why did you choose AI & Data Science?

I was interested in solving business problems using data. During my engineering, I learned SQL, Power BI, Python, Data Analytics, Cloud Computing, and Big Data, which motivated me to build a career in Data Engineering and Analytics.

3. Why Data Engineering instead of AI and ML?

I enjoy ETL/ELT Process, building data pipelines, and preparing data for analytics and AI/ML Development Teams. I like working on data infrastructure more than developing machine learning models.

4. What have you learned during your internship?

During my Internship, I got exposure about project workflows, project architecture, work management, and the process of solving real business problems by collaborating with experienced Data Professionals and business teams. This internship helped me to understand how different teams work together to deliver reliable, data-driven solutions.

5. Why Tata Motors instead of Others?

I choose Tata Motors, because it is one of the most respected global manufacturing companies, known for its innovation, large client base, and strong focus on automotive transformation.

It gave me exposure to enterprise analytics, scale data pipelines, and solving approach of real-world business problems.

6. Why are you looking for a new opportunity?

I want to take on a full-time role, where I can work on larger data projects, continue learning, grow professionally, and contribute to the growth of organization.

Project Based Questions

7. Explain the Project / Which Part of Project are you most proud of? (Fleet Edge)

During my Internship I have worked and assisted on multiple projects. One of the major projects is Fleet Edge, a commercial vehicle analytics platform used to monitor vehicle performance using IoT Sensors telemetry and CAN (Control Access Network) data.

As a Data Analytics Intern, I worked with the Central Analytics team where my responsibilities were to validate automated data pipelines, monitor daily pipeline executions, analyze logs whenever a pipeline failed, and report the root causes to the senior Data Engineering team.

The data was processed using Big Data frameworks such as PySpark and Spark SQL and stored in Azure Cloud, particularly in Azure Data Lake Storage in Parquet file format.

The processed data was later used for reporting and predictive modelling to measure fuel efficiency, trip analysis, load capacity, and users driving behaviour.

8. What was your exact contribution? (Fleet Edge)

I worked under Senior Data Engineers and Assisted them in maintaining reliable data pipelines.

My responsibilities included:

- Monitoring daily automated pipelines
- Validating processed data
- Checking pipeline execution logs
- Performing root cause analysis when pipelines failed
- Processing IoT sensors and CAN data
- Verifying data quality before available to reporting and predictive modelling teams

9. What kind of data was coming into the pipeline?

The data pipeline received IoT telemetry sensors and CAN data generated from commercial vehicles.

The data included:

- Vehicle ID, Customer ID, Trip Details, Distance Travelled, Fuel Consumption, Load Capacity, GPS Location, Timestamp, etc.

10. Why Particular Tech Stack used? (Fleet Edge)

The Fleet Edge project processes large volumes of IoT telemetry sensors and CAN data generated by commercial vehicles every day, so the technology stack was chosen based on scalability, performance, and integration.

(PySpark, Spark SQL, Azure Cloud, Azure Databricks, Fabric, Parquet Files)

11. Why Parquet File format is used instead of others? (Fleet Edge)

We used the Parquet file format because, it is a columnar storage format that provides better compression, faster query performance, and lower storage usage compared to CSV or JSON.

It also integrates well with PySpark and Azure cloud (Azure Data Lake Storage), make it ideal for big data processing and analytics.

12. Why Spark instead of Python? (Fleet Edge)

Python works well for small to medium datasets on a single machine, but Spark is designed for big data processing. Spark distributes the workload across multiple machines, making data processing much faster, scalable.

13. What Validations did you perform? / How to ensure data Quality? (Fleet Edge)

- Null & Missing Values, Duplicate Records, Schema Validation, Data Type Validation, Record Count Validation, Mandatory Fields Validation, Source vs Target Data Validation, etc.

14. What challenges did you face? (Fleet Edge)

Pipeline failures due to data quality issues. I analyzed logs, identified the root causes, and reported issues for resolution and solved it by discussed with Senior Data Engineers.

15. What kind of Pipeline failures did you observe?

- Missing or Delayed input files / Schema changes / Null values / Data type mismatch / Duplicate records / Corrupted files / Pipeline execution failure / Log Analysis & Root Cause Analysis, etc.

16. Describe a difficult problem. (Fleet Edge)

Data pipelines failed because of inconsistent data. For that, I checked logs, identified the issue of failure, validated the corrected data, and ensured successful execution.

17. How to handle Schema Evaluation? (For All Projects)

I validated column names, data types, mandatory fields, null values, and schema consistency against the expected schema. If any mismatch was found, I analyzed the logs, identified the root cause, and reported the issue before processing continued.

18. What did Business benefit? (For All Projects)

The project improved data quality by providing accurate and reliable data for reporting and analytics. It reduced manual effort through automation, enabled faster reporting, improved operational efficiency, and helped business teams make better data-driven decisions.

19. How do you handle deadlines?

I prioritize tasks, break down the problem into smaller steps, Communicate regularly with my team, and complete work according to deadlines.

Cross Projects Questions

20. Which Project was technically challenging?

Fleet Edge was the most challenging project because it involved large-scale IoT Sensors and CAN data processing, pipeline validation, troubleshoot the production pipeline failures, and ensure high data quality in a real-world environment.

21. Which Technology did you learn most?

During my internship, I learned the most about **Spark SQL**. I used it to process and validate large volumes of IoT telemetry data, perform data transformations, and support automated data pipelines. It helped me understand how big data is processed efficiently in a production environment.

22. What monitoring and alerting mechanisms have you implemented?

I monitored pipeline execution status and logs, investigated failure alerts, identified root causes, and reported issues to ensure smooth pipeline operations.

23. Did you build the pipeline yourself? (Important Question)

No. The production pipelines were already developed by the senior Data Engineering team. My role was to validate daily pipeline executions, monitor failures, analyze logs for root cause analysis, process and verify data using PySpark and Spark SQL, and ensure high data quality before it was used for reporting and predictive modeling.

Behavioral Questions

24. Why should we hire you? / Why are you interested in this role?

You should hire me because I believe that, I am a good fit for this role, because my skills and internship experience closely match the requirements of the job. I have hands-on experience working on real-world data projects and understand how to apply skills to solve business problems.

I am a quick learner, adaptable, and eager to contribute to team while continuously improving my skills. I am confident that I can add value to the organization and grow along with it.

25. Why should we prefer you over other candidates?

I may not know the strengths of others, but I bring dedication, quick learning, and teamwork. I am confident that I will adapt fast and contribute effectively.

26. What are your strengths and Weakness?

Problem-solving, Willingness to learn, teamwork, and attention to detail.

For the Weakness, I sometimes overthink before starting something new, but I have been working on it by taking small steps and staying consistent, which I am actively improving.

27. Where do you see yourself in five years? / What are your career Goals?

In 5 years, I see myself as a skilled and responsible Data Professional, leading data end to end data projects and contributing more value to company.

28. If we don't hire you, what will you do?

I will keep improving my skills and prepare be er for next me. I will also look for other opportunities where I can contribute and grow.

29. What do you expect from the company?

I expect a good learning environment, guidance from seniors, and opportunities to grow along with the company.

30. Why did you choose this company? / Why do you want to join this company?

I admire the work your company is doing in [industry/field: "IT/ retail"]. I like how you focus on innovation. As a fresher, I feel this role gives me a chance to contribute my skills, learn new technologies and also learn from your experienced team.

31. Are you willing to relocate/work in shifts?

Yes, I am flexible with relocation and shifts if the role requires it.

32. What do you know about our company?

I know your company is one of the leading organizations in the IT industry, known for delivering its innovative solutions and strong global presence.

I really admire your company focus on learning, and digital transformations, and also the way you encourage young professionals to grow through challenging projects and that makes me interested in being part of it.

33. What are your Salary expectations?

For me, my main goal is to start my career in a company where I can learn and contribute effectively.

Could you please share the standard salary budget/range that company has set for this role, so that I can respond it accordingly.

34. How much do you expect as a fresher?

As a fresher, I don't have a fixed number in mind. I trust your company to offer a fair package based on industry standards.

I am open to the company's standard package for this role. For me, right now learning and growth are more important.

35. Do you have any questions for us?

Yes. I'd like to know:

- What technologies does the team use daily? And What kind of projects would I work on?
- Do you have any feedback on my technical knowledge performance today? I would like to know that what I should need to improve.

Final 30-Second Closing Line

Thank you for giving me the opportunity to interview with you today. I really liked our discussion and company's culture. I am excited about the opportunity to be part of your team, contribute to the company, and grow professionally.

Please let me know if there is any other information you would like from my side, and I look forward to hearing from you about the next process.